

Reading Between the Lines: LLM-Powered Topic Modelling and Graph-Based Insights from Research Abstracts

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Abstract. The exponential growth of research literature across domains has brought along with it difficulties in successfully analyzing big corpora to identify upcoming themes, underlying relations, and evolving trends. This work proposes a new framework that merges Large Language Models (LLMs) with cutting-edge topic modeling and graph analysis to address these challenges. Semantic embeddings are created and optimized with Uniform Manifold Approximation and Projection (UMAP) and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) for improved topic relevance. After topic assignment, keyword extraction is performed using KeyBERT and further optimized using Maximal Marginal Relevance (MMR). The LLM aggregates representative documents in order to enhance the interpretability of topics, after which they are converted into graph models to reveal centrality features and influence within research themes. This dual strategy offers both scalability and interpretability and outperforms previous strategies in the identification of sophisticated patterns and offering actionable insights. Future research could focus on dynamic topic tracking to uncover temporal trends, highlighting emerging and declining research areas.

Keywords: Topic Modelling, BERTopic, LLama2, Graph Analytics, LLM

1 Introduction

Over the past decade, the exponential growth in research output has made the ability to analyze and synthesize large corpora essential for uncovering new themes, hidden patterns, and evolving trends. While traditional topic modeling approaches are useful, they often fall short in capturing the complexity and interdisciplinary nature of contemporary research datasets. To address these challenges, this paper proposes a novel framework that integrates Large Language Models (LLMs) with advanced topic modeling and graph analytics.

Our approach uses LLaMA 2 to generate coherent, contextually relevant topic labels, which are then refined with BERTopic—a framework that applies HDBSCAN for clustering and UMAP for dimensionality reduction. Keywords are extracted using KeyBERT and further refined with Maximal Marginal Relevance

(MMR) to balance diversity and relevance in topic representation. A key contribution of our method is the transformation of topic clusters into graph structures, enabling the identification of hidden relationships, centrality, and influence among research themes. This hybrid approach enhances interpretability and reveals interdisciplinary connections often overlooked by conventional methods.

By combining LLM-based topic generation with graph analysis, our system offers a scalable and interpretable method for investigating large-scale research literature. It equips researchers with actionable insights into the broader research landscape, helping identify emerging trends and inform future research directions. Our findings demonstrate the effectiveness of the dual approach in uncovering nuanced patterns and advancing the analysis of complex research corpora.

2 Literature Survey

The applications of advanced machine and deep learning techniques such as LLMs and graph-based analytics have recently experienced tremendous growth. This section highlights key studies that have advanced topic modeling methodologies. The selected papers below provide foundational insights into the approaches and challenges of leveraging these techniques.

Yida Mu (2024) (1): This paper explores using Large Language Models (LLMs) as an alternative to traditional methods like LDA, overcoming limitations in semantic understanding and topic granularity. The authors propose a prompt-based framework for LLM-driven topic extraction, evaluate its effectiveness, and introduce constraints and post-processing techniques to improve topic quality.

Han Wang (2023) (2): This paper combines LLMs with a novel guiding vector to develop PromptTopic, a sentence-level model that eliminates the need for manual tuning and operates at a paragraph-level semantic granularity.

Schneider (2024) (3): This work introduces FT-Topic, an unsupervised fine-tuning method for LLM-based topic modeling that automatically builds a training dataset from sentence groups. The fine-tuned encoder enhances topic modeling, demonstrated through SenClu, which enables efficient inference, hard topic assignments, and user-defined topic distributions.

Rana Nassif (2020) (6): This study combines Hidden Markov Models (HMMs) with deep learning techniques (CNN and LSTM) and word embeddings (Word2Vec and GloVe) for supervised document classification. The proposed hybrid model outperforms existing methods by capturing both word order and semantic meaning.

Dazhong Shen (2021) (11): This work introduces the Graph Neural Topic Model (GNTM), which represents documents through sparsely annotated directed graphs capturing word dependencies, thereby addressing key limitations of traditional Bag-of-Words models.

Greg Bacon (2024) (14): This paper employs LLM embeddings combined with dimensionality reduction, clustering, and heuristics to predict the suitability

of research topics. The method identifies significant topics structured around the content of research items.

3 Methodology

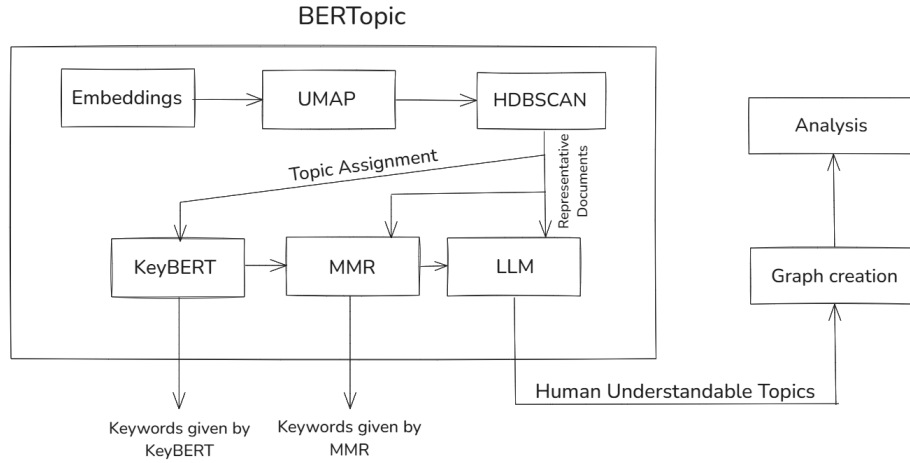


Fig. 1: Topic Generation Pipeline

Fig. 1 outlines the workflow of BERTopic and the results it produces. The final results generated by BERTopic, combined with the LLM, are then used to construct a graph.

3.1 Data Collection and Preparation

The dataset used in this project consists of research papers with parameters such as `abstract`, `authors`, `n-citation`, `references`, `title`, `venue`, `year`, and `id`. The original dataset contained approximately 0.1 million rows. We randomly sampled 25,000 rows using simple random sampling. For our analysis, we focused on the `abstract` and `title` columns, as they were the most relevant to our study.

3.2 Generating Text Embeddings

After extracting the abstracts, we generated text embeddings using the `BAAI/bge-small-en` model, a compact English embedding model developed by the Beijing Academy of Artificial Intelligence (BAAI).

3.3 Optimizing Embedding Space

To enhance the clustering of similar embeddings in the embedding space, we reduced their dimensionality using UMAP (Uniform Manifold Approximation and Projection), a non-linear dimensionality reduction technique that preserves both local and global structures. UMAP is particularly effective for high-dimensional data visualization and clustering.

3.4 Clustering and Topic Assignment

The reduced embeddings were fed into HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise), a clustering algorithm that identifies dense regions in the data. It groups semantically similar abstracts into clusters and assigns a topic label to each cluster. Data points identified as outliers by the model are labeled as -1 .

3.5 Extracting Representative Keywords for Topics

Documents assigned to the same topic cluster are grouped together, and the most representative documents within each cluster are identified. Keyword extraction is then performed using a BERT-based method, which evaluates the relevance of each word or phrase based on its semantic similarity to the topic representation. The resulting keywords act as concise and meaningful descriptors for each topic, enhancing the interpretability and understanding of the clustered content.

3.6 Enhancing Keyword Selection

To improve keyword diversity and reduce redundancy, Maximal Marginal Relevance (MMR) is applied after initial keyword extraction. This method balances relevance and diversity by considering both the representative documents identified by HDBSCAN and the keywords extracted via KeyBERT. While KeyBERT selects keywords based on their semantic similarity to topic-related documents, MMR iteratively selects terms that are not only relevant but also distinct from previously chosen ones. By penalizing semantic overlap, MMR produces a more diverse and comprehensive keyword set, ultimately enhancing the interpretability of each topic.

3.7 Generating Refined Topic Names

After topic generation and keyword refinement via MMR, BERTopic typically treats the resulting labels as final. In our framework, however, we incorporate LLaMA2 to further refine the topic names. We prompt LLaMA2 using both the MMR-refined keywords and the corresponding representative documents, allowing it to produce concise, human-readable topic names. To maintain consistency and control over the output structure, we employ a one-shot prompting strategy, which guides the model to generate names that align with our desired format and level of abstraction.

3.8 Building Graph for the Topics

After obtaining the topics, we constructed a graph based on the similarity between them. Cosine similarity was used as the similarity metric, with a threshold value of 0.2. Edges were created between topic nodes only if their similarity exceeded this threshold, allowing us to visualize and analyze inter-topic relationships effectively.

3.9 Graph Analytics

We extracted valuable insights from the graph using the following algorithms:

- **Degree Centrality:** Measures the importance of a node based on the number of edges connected to it. Nodes with higher degree centrality are more connected and influential within the network.

$$C_{\text{degree}}(G) = \frac{\sum_{i=1}^n (C_{\text{degree}}(v_i)_{\max} - C_{\text{degree}}(v_i))}{(n-1)(n-2)}$$

- **Community Detection:** The Louvain algorithm is used for community detection, identifying groups of closely connected nodes within a large network by optimizing the modularity score. This method effectively detects communities where internal connections are denser than connections to other communities, helping to reveal distinct communities within the graph. Modularity maximization is used to detect and visualize these communities.

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

- **Subgraph Generation:** Extracts a subset of the graph by selecting a specific topic and retrieving all related topics for analysis. This helps analyze topic-specific interactions and dependencies.

$$S(H) = \sum_{i,j \in H} A_{ij}$$

- **Betweenness Centrality:** Determines the significance of an edge by computing how often it appears in the shortest paths between other nodes. Higher values indicate that the edge plays a crucial role in network connectivity. The top 5 edges with the highest betweenness centrality are identified.

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

- **K-core:** Identifies the most well-connected nodes by determining the highest-order core to which each node belongs. Nodes with the highest core number form the core structure of the graph, playing a central role in the network. The key topics in the highest core are highlighted.

$$V_k = \{v \in V \mid d_H(v) \geq k\}$$

KeyBERT
['algorithms', 'networks', 'research', 'algorithm', 'models', 'network', 'model', 'simulation', 'users', 'data']
['throughput', 'networks', 'network', 'nodes', 'algorithm', 'relay', 'mimo', 'channels', 'allocation', 'routing']
['segmentation', 'camera', 'model', 'video', 'image', 'estimation', 'images', '3d', 'visual', 'detection']
['language', 'speech', 'corpus', 'model', 'models', 'learning', 'acoustic', 'search', 'translation', 'semantic']
['modeling', 'software', 'models', 'model', 'developers', 'implementation', 'systems', 'research', 'development', 'innovation']
['fmri', 'cortical', 'cortex', 'brain', 'neural', 'mri', 'eeg', 'neurons', 'model', 'models']
['bioinformatics', 'microarray', 'genes', 'genomic', 'genome', 'proteins', 'gene', 'rna', 'genomes', 'biological']
['circuits', 'adc', 'cmos', 'circuit', 'transistor', 'analog', 'faults', 'chip', 'memory', 'technology']
['robotics', 'robots', 'manipulators', 'robotic', 'manipulator', 'robot', 'kinematic', 'dynamics', 'trajectory', 'trajectories']
['regression', 'bayesian', 'models', 'model', 'classifiers', 'datasets', 'algorithms', 'classifier', 'clustering', 'algorithm']

Table 1: KeyBERT Topics

4 Results

Our work combines Large Language Models with advanced topic modeling and graph analytics, showcasing the potential of this hybrid approach for analyzing extensive research literature. By using LLaMA 2 to generate high-dimensional semantic embeddings, we were able to recover coherent and meaningful themes from a dataset of 25,000 research papers. We used BERTopic for robust topic modeling, further enriched with techniques such as UMAP for dimensionality reduction and Maximum Marginal Relevance for improving topic relevance, which provided us with well-separated and interpretable clusters of topics and subtopics. It provided insightful insights into the trends and latency in the specific research areas addressed.

Fig 2 highlights the structured separation of topics, ranging from "Multi Channel Network Optimization" to "Graph Theory and Paradox," which shows that the model can create well-defined topic groups. Fig 3. displays the different topic cluster labels generated by LLaMa-2. Table 1 presents keywords extracted using KeyBERT from the reduced embeddings, Table 2 showcases topic labels generated by LLaMA-2. Table 3 shows the results obtained through MMR from the representative documents. These tables represent output for 10 sample values, and the entire process happens within the BERTopic framework.

Graph analytics added significant depth to our results. In our case, Fig. 4. shows topic clusters transformed into graph structures and allows analysis of



Fig. 2: Clustering Visualization

- Multi-Channel Network Optimization
- Image Processing
- Speech Recognition and Language Modeling
- Software Development Process Improvement
- Medical Imaging and Brain Research
- Protein-Protein Interactions and Gene Expression Analysis
- Power-Efficient Design Automation
- Robot Control and Motion Planning
- Computational Efficiency in Bayesian Analysis
- Memory Management in Modern Processors
- Graph theory and combinatorics
- Remote Sensing and Land Surface Parameters
- Game Theory Paradox
- Signal Processing and Estimation
- Learning and Education
- Control and Stability Analysis for Nonlinear Systems
- Numerical Methods for Solving Nonlinear Equations
- Logical Semantics and Formal Verification
- Optimization in Dynamic Environments
- Cryptographic Security Protocols
- XML Query Optimization
- Power Systems Control and Optimization
- Healthcare Information Systems
- Virtual Interaction and Presence in Virtual Reality
- Resource Management in Cloud Computing

Fig. 3: Topic Labels

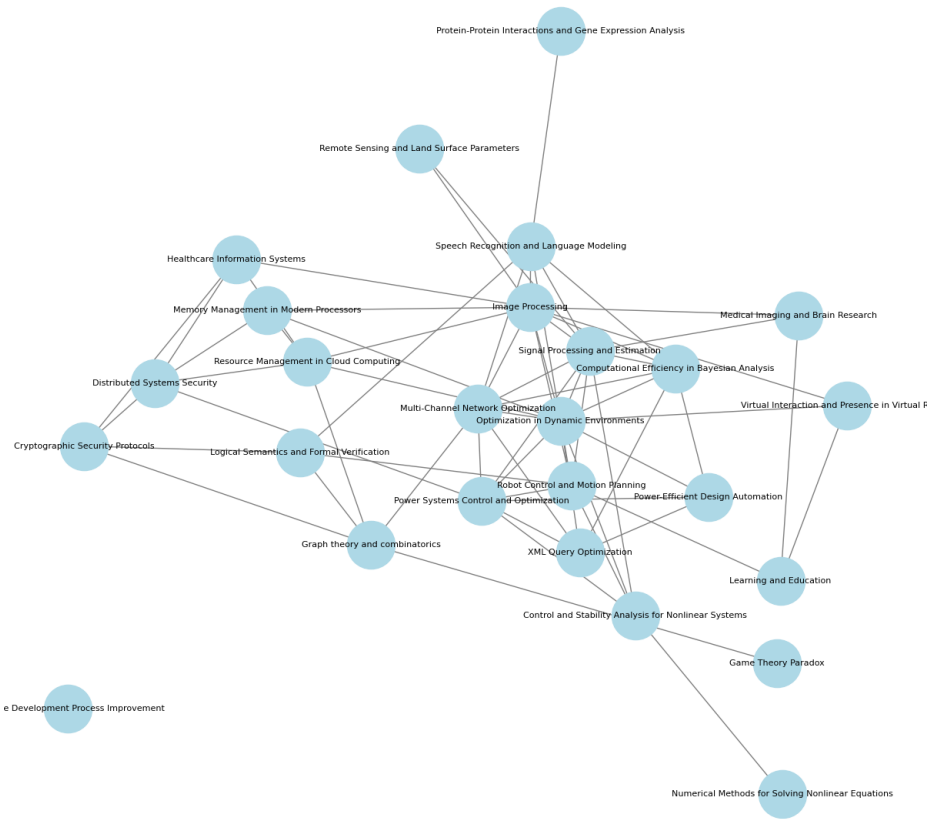


Fig. 4: Graph Visualization

Llama2 Topics
Distributed Systems Security
Multi-Channel Network Optimization
Image Processing
Speech Recognition and Language Modeling
Software Development Process Improvement
Medical Imaging and Brain Research
Protein-Protein Interactions and Gene Expression Analysis
Power-Efficient Design Automation
Robot Control and Motion Planning
Computational Efficiency in Bayesian Analysis

Table 2: Llama2 Topics

relationships, centrality, and influence of different research topics. Fig. 5. illustrates research communities, where the nodes represent different research areas and are connected. Different clusters, coloured differently, show possible synergies and knowledge transfer paths across disciplines. Table 4 represents the different research communities identified after applying the Louvain community detection algorithm. In Fig 6. the subgraph community of "Image Processing" revealed new hidden connections that otherwise would seem to be entirely unrelated domains.

These integration tools therefore have laid down a foundation where promising areas for exploration can be identified by unveiling actionable insights into the discovery, and guiding other disciplines' further research directions.

tabularx array ragged2e

The topics generated by Llama-2 were compared against the paper titles using cosine similarity. Table 5 summarizes the resulting similarity scores.

5 Conclusion

In this study, we successfully demonstrated the integration of Large Language Models (LLMs) with advanced clustering techniques and graph analytics to analyze a corpus of literature. After generating topic labels from LLaMA 2 and BERTopic at the topic modeling level, coherent topics and subtopics can be derived. Graph analytics has enriched our analysis by enabling the identification of relationships between topics and communities. Converting topic clusters into graph structures results in better insights into centrality and influence among the topics within a broader landscape. This dual approach has proven to be scalable and interpretable, thus superior to traditional methods. Overall, the proposed framework offers an efficient method for analyzing research, uncovering nuanced patterns, and providing actionable insights for researchers.

MMR Topics
['of', 'to', 'for', 'as', 'an', 'data', 'paper', 'system', 'model', 'systems']
['channel', 'network', 'performance', 'networks', 'wireless', 'scheme', 'paper', 'can', 'rate', 'results']
['image', 'to', 'images', 'method', 'video', 'an', 'from', 'as', 'algorithm', 'motion']
['speech', 'word', 'as', 'language', 'web', 'model', 'text', 'recognition', 'paper', 'features']
['software', 'process', 'business', 'model', 'development', 'knowledge', 'paper', 'requirements', 'approach', 'systems']
['of', 'brain', 'for', 'from', 'images', 'using', 'results', 'imaging', 'data', 'model']
['protein', 'gene', 'data', 'genes', 'from', 'proteins', 'genome', 'expression', 'biological', 'database']
['test', 'circuit', 'circuits', 'fault', 'cmos', 'voltage', 'chip', 'at', 'can', 'faults']
['robot', 'control', 'robots', 'motion', 'planning', 'method', 'force', 'manipulator', 'path', 'mobile']
['data', 'learning', 'clustering', 'algorithm', 'classification', 'methods', 'feature', 'method', 'algorithms', 'model']

Table 3: MMR Topics

6 Future work

Future work can focus on several aspects to enhance the applicability of the proposed framework. Developing methodologies for dynamic topic tracking could provide insights into the temporal evolution of research areas, enabling the identification of emerging fields and declining trends. Scalability to larger corpora is another significant path forward, where extending the framework to support millions of papers would require optimizations in the generation of embeddings, clustering, and graph analytics. The inclusion of more powerful techniques for dimensionality reduction, such as variants of t-SNE or deep generative models, can reveal subtler topic structures and improve the quality of visualization. More importantly, benchmarking the proposed hybrid approach against state-of-the-art methods, such as prompt-based topic modeling techniques and graph neural networks, is essential to validate its efficacy. Its adaptability can be improved by further developing a domain-specific application framework, such as for healthcare or AI, tuning the embeddings and clustering algorithms according to discipline-specific characteristics, and expanding graph analytics by integrating other structured knowledge graphs, such as OpenCitations or Semantic Scholar graphs, to uncover richer contextual relationships and interdisciplinary linkages. Finally, automating recommendations for relevant literature or collaborations based on identified trends would add significant practical value for both academic and industry researchers.

Community Number	Research Topics
1	Distributed Systems Security, Memory Management in Modern Processors, Graph theory and combinatorics, Game Theory Paradox, Cryptographic Security Protocols, Healthcare Information Systems, Resource Management in Cloud Computing
2	Multi-Channel Network Optimization, Power-Efficient Design Automation, Computational Efficiency in Bayesian Analysis, Optimization in Dynamic Environments, XML Query Optimization, Power Systems Control and Optimization
3	Image Processing, Speech Recognition and Language Modeling, Medical Imaging and Brain Research, Protein-Protein Interactions and Gene Expression Analysis, Robot Control and Motion Planning, Remote Sensing and Land Surface Parameters, Signal Processing and Estimation, Learning and Education, Logical Semantics and Formal Verification, Virtual Interaction and Presence in Virtual Reality
4	Software Development Process Improvement
5	Control and Stability Analysis for Nonlinear Systems, Numerical Methods for Solving Nonlinear Equations

Table 4: Research Communities

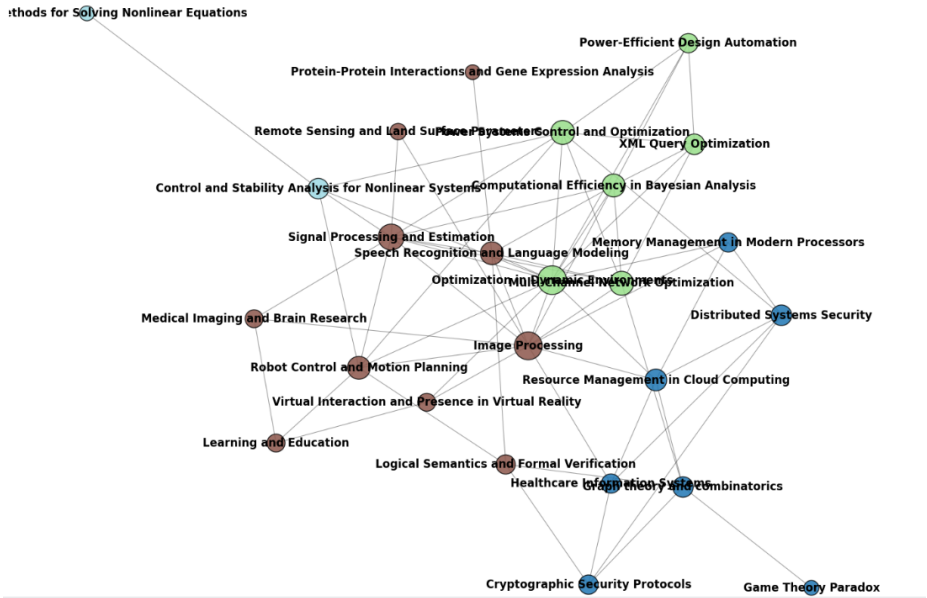


Fig. 5: Community Visualization

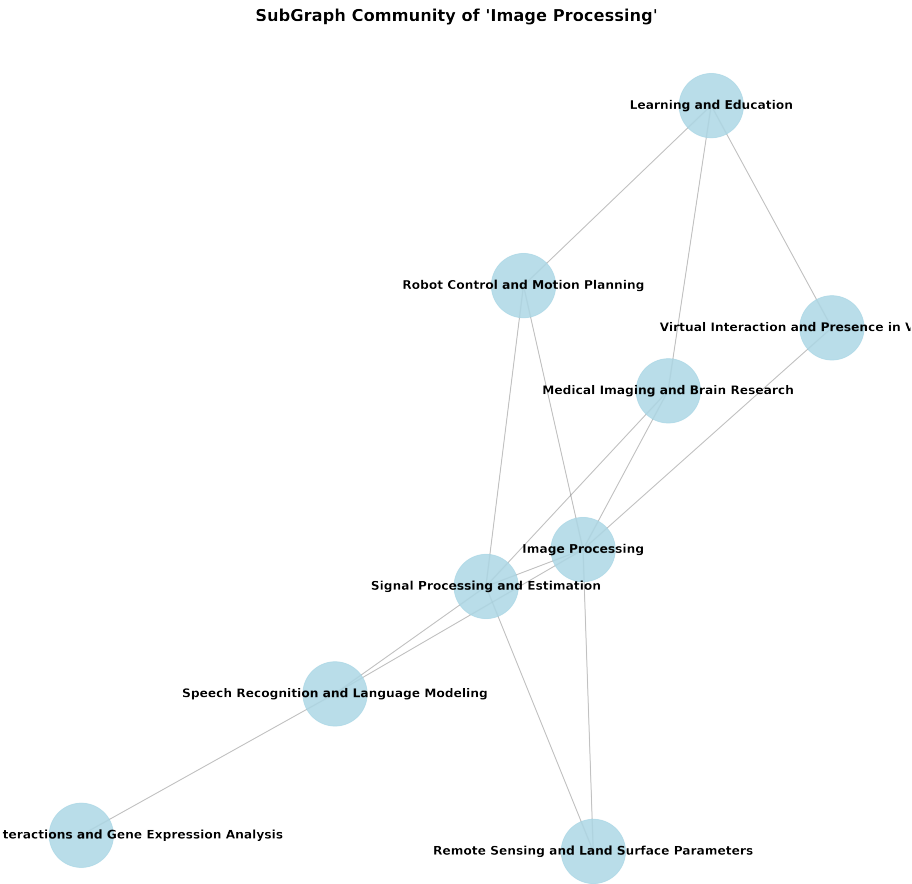


Fig. 6: SubGraph Visualization

Paper Title	Generated Topic	Cosine Similarity Score
Recursive XML schemas, recursive XML queries, and relational storage: XML-to-SQL query translation	XML Query Optimization	0.656
Elimination graphs	Graph theory and combinatorics	0.602
Subspace Approach for Fast and Accurate Single-Tone Frequency Estimation	Signal Processing and Estimation	0.581
Translation and rotation of virtual objects in Augmented Reality: A comparison of interaction devices	Virtual Interaction and Presence in Virtual Reality	0.573
Virtual Hypervisor: Enabling fair and economical resource partitioning in cloud environments	Resource Management in Cloud Computing	0.525

Table 5: Cosine Similarity Scores for Paper Topics

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